

Residual certificates for the Erdős–Straus conjecture: exact solvability criteria, a sieve bound, and computations to 10^{10}

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August 2026 — draft

Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/a + 1/b + 1/c$ is solvable in positive integers for every $n \geq 2$. The open cases concentrate on primes in the six residue classes $\{1, 121, 169, 289, 361, 529\} \bmod 840$ (*hard primes*). We study the *residual method*: for $R \equiv 3 \pmod{4}$ set $a = (p + R)/4$, $m = pa$; a solution with first denominator a exists if and only if some divisor $k \mid m^2$ satisfies $k \equiv -m \pmod{R}$. Our main theoretical results are: (i) *exact* solvability criteria for the residuals $R = 3, 7$, and 11 at hard primes, the latter two established by machine-verified finite case analyses; (ii) a meta-theorem showing such a finite exact criterion exists and is computable for every fixed R ; (iii) an unconditional sieve bound: the number of hard primes $p \leq x$ solved by neither $R = 3$ nor $R = 7$ is $O(x/(\log x)^2)$, and with $R = 11$ included, $O(x/(\log x)^{5/2})$; (iv) a general density reduction: for every A there is a finite residual list whose joint exceptional set has relative density $O((\log x)^{-A})$ among hard primes. Computationally, we exhibit verified residual certificates for all 14,215,707 hard primes below 10^{10} : the maximal minimal residual is 107, attained at the single prime 8,803,369 and unchanged from 5×10^7 to 10^{10} ; a fixed list of 18 residuals covers every hard prime below 10^9 . The measured density of the joint $\{3, 7\}$ -exceptional set is $\approx 5.17/\log x$, matching the sieve exponent exactly.

1 Introduction

Erdős and Straus conjectured that for every integer $n \geq 2$ the equation

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \tag{1}$$

has a solution in positive integers a, b, c . By multiplicativity it suffices to treat prime n , and classical polynomial identities settle every prime outside the six residue classes

$$\mathcal{H} \bmod 840 = \{1, 121, 169, 289, 361, 529\},$$

the squares of units modulo 840 (Mordell [4]). We call primes in these classes *hard primes*. Direct verification has confirmed the conjecture far beyond any range considered here (to 10^{14} by Swett, to 10^{17} by Salez [5], and further in recent work¹); the structure of the solution count was analyzed by Elsholtz and Tao [1]. Verification, however, exhibits no mechanism; the aim here is a quantitative structural reduction.

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¹Verification to 10^{18} is reported in recent literature; the present paper is logically independent of that bound.

1.1 The residual method

Fix a hard prime p . All six classes satisfy

$$(H1) \ p \equiv 1 \pmod{8}, \quad (H2) \ p \equiv 1 \pmod{3}, \quad (H3) \ p \bmod 7 \in \{1, 2, 4\}. \quad (2)$$

For an *admissible residual* $R \equiv 3 \pmod{4}$ put

$$a = \frac{p+R}{4} \in \mathbb{Z}, \quad m = pa.$$

Substituting a into (1) and clearing denominators shows that a solution with first denominator a exists if and only if

$$\exists k \mid m^2 \quad \text{with} \quad k \equiv -m \pmod{R}, \quad (3)$$

in which case $b = (k+m)/R$ and $c = (m^2/k+m)/R$ are positive integers solving (1). We write $R_{\min}(p)$ for the least admissible R for which (3) holds. Two elementary congruences drive everything below. From $4a = p+R$,

$$a \equiv 4^{-1}p, \quad m \equiv 4^{-1}p^2 \pmod{R}, \quad (4)$$

and since $\gcd(p, R) = 1$ (as $p > R$ is prime), also $\gcd(a, R) = \gcd(m, R) = 1$: the setting is always a unit computation modulo R . By (4), m is a *square* modulo every prime $r \mid R$; hence for every prime $r \mid R$ with $r \equiv 3 \pmod{4}$ (at least one exists, since $R \equiv 3 \pmod{4}$),

$$\left(\frac{-m}{r}\right) = \left(\frac{-1}{r}\right) = -1 : \quad (5)$$

the target class $-m$ is a quadratic non-residue mod r .

1.2 Main results

Theorem 1.1 ($R = 3$; exact). *Let $p \equiv 1 \pmod{4}$ with $3 \nmid p$, and $a = (p+3)/4$. Then (3) holds for $R = 3$ if and only if a has a prime factor $q \equiv 2 \pmod{3}$. When it does, $k = q$ is a certificate.*

Theorem 1.2 ($R = 7$, hard primes; exact). *Let p be a hard prime and $a = (p+7)/4$. Then (3) holds for $R = 7$ if and only if a has a prime factor that is a quadratic non-residue modulo 7.*

Theorem 1.3 ($R = 11$, hard primes; exact). *Let p be a hard prime and $a = (p+11)/4$; note $3 \mid a$, $2 \nmid a$, $7 \nmid a$ automatically. Then (3) fails for $R = 11$ if and only if either*

- (a) *every prime factor of a is a quadratic residue mod 11; or*
- (b) *$v_3(a) = 1$, every other prime factor of a is $\equiv 1 \pmod{11}$ except a non-residue part of exactly one of the shapes*

$$\{q \equiv 2\}, \quad \{q \equiv 6\}, \quad \{q \equiv 2\} \cup \{q' \equiv 6\} \pmod{11},$$

each with multiplicity one, paired respectively with $p \equiv 2, 6, 1 \pmod{11}$.

Theorems 1.2 and 1.3 are established by finite, machine-verified case analyses (Section 3); the reduction to a finite check is itself a theorem:

Theorem 1.4 (meta-theorem). *For every fixed prime $R \equiv 3 \pmod{4}$, the validity of (3) at a hard prime p depends only on (i) the multiset of classes mod R of the prime factors of $a = (p + R)/4$, with multiplicities capped at $\lceil (R - 1)/2 \rceil$, and (ii) $p \pmod{R}$. Consequently an exact solvability criterion exists for every R and is computable by finite enumeration.*

The anchor result converts the exact criteria into an unconditional bound.

Theorem 1.5 (sieve bound). *The number of hard primes $p \leq x$ with $R_{\min}(p) \geq 11$ (that is, solved by neither $R = 3$ nor $R = 7$) is $O(x/(\log x)^2)$. With Theorem 1.3 the count of hard primes with $R_{\min}(p) > 11$ is $O(x/(\log x)^{5/2})$. In particular these exceptional sets have relative density $O((\log x)^{-1})$, resp. $O((\log x)^{-3/2})$, among hard primes.*

Theorem 1.6 (density reduction). *For every $A > 0$ there is a finite admissible set $S(A)$ such that the number of hard primes $p \leq x$ for which every residual in $S(A)$ fails is $O_A(x/(\log x)^{1+A})$.*

Computations (Section 5) provide verified certificates for all hard primes below 10^{10} and calibrate every constant in sight.

2 The obstruction framework

Proposition 2.1 (character obstruction). *Let $r \mid R$ be prime with $r \equiv 3 \pmod{4}$. If every prime factor q of $m = pa$ satisfies $\left(\frac{q}{r}\right) = +1$, then (3) fails for R .*

Proof. Every divisor k of m^2 is a product of prime factors of m , so $\left(\frac{k}{r}\right) = +1$. A certificate requires $k \equiv -m \pmod{r}$, whence $\left(\frac{k}{r}\right) = \left(\frac{-m}{r}\right) = -1$ by (5) — impossible. $\square$

We call failures produced by Proposition 2.1 *Type I*; failures where m possesses a non-residue prime factor modulo every relevant r , yet no divisor of m^2 attains the exact class $-m \pmod{R}$, are *Type II*.

Proposition 2.2 (guaranteed-success classes). *Let $t \equiv -4^{-1}p^2 \pmod{R}$. If a has a prime factor q with $q \equiv tp^{-i} \pmod{R}$ for some $i \in \{0, 1, 2\}$, then $k = qp^i$ is a certificate. In particular, since $tp^{-2} \equiv -4^{-1} \pmod{R}$, any prime factor of a in the fixed class $-4^{-1} \pmod{R}$ certifies success regardless of p .*

Proof. $k = qp^i$ divides m^2 (as $q \mid a$ and $i \leq 2$) and lies in class $t \equiv -m$. Integrality of b, c is automatic from $k \equiv -m$ and $m^2/k \equiv m^2(-m)^{-1} \equiv -m \pmod{R}$. $\square$

The contrapositive of Proposition 2.2 is the sieve input for Theorem 1.6: failure at R forces $(p + R)/4$ to avoid prime factors in up to three explicit classes modulo R .

3 Exact criteria

3.1 Proof of Theorem 1.1

Modulo 3 the unit group has order 2, so quadratic character equals class. Since $p^2 \equiv 1 \pmod{3}$, (4) gives $m \equiv 1$ and the target is $-m \equiv 2 \pmod{3}$; also $3 \nmid a$. *Necessity:* if every prime factor of a is $\equiv 1 \pmod{3}$, then $a \equiv 1 \pmod{3}$, hence $p = 4a - 3 \equiv 1 \pmod{3}$ as well, so every prime factor of $m = pa$ is $\equiv 1 \pmod{3}$ and every divisor of m^2 is $\equiv 1 \pmod{3}$: class 2 is never attained. *Sufficiency:* $k = q \equiv 2 \pmod{3}$ divides m^2 , and the recovery formulae give integers as in Proposition 2.2. $\square$

3.2 Proof of Theorem 1.4

A certificate is a divisor $k \mid m^2$ in class $-m \bmod R$. The set of classes of divisors of m^2 is determined by the classes of the prime factors of $m = pa$ together with their multiplicities, each factor being usable with exponent at most twice its multiplicity; exponents are only meaningful modulo $\text{ord}_R(\text{class}) \leq R - 1$, so multiplicities beyond $\lceil (R - 1)/2 \rceil$ are equivalent to that cap. The target class and the consistency relation $\prod(\text{classes}) \equiv a \equiv 4^{-1}p \pmod{R}$ are functions of $p \bmod R$ by (4). Hence solvability is a function of the capped class multiset and $p \bmod R$, a finite amount of data. $\square$

3.3 Proof of Theorem 1.2

($\Rightarrow$) is Proposition 2.1 with $r = R = 7$. ($\Leftarrow$) reduces to a finite verification via Theorem 1.4. Write $(\mathbb{Z}/7)^* \cong \mathbb{Z}/6$ in discrete-log coordinates. Three structural constraints hold at hard primes:

- (i) $p \bmod 7 \in \{1, 2, 4\}$, by (H3);
- (ii) $2 \mid a$: by (H1), $p + 7 \equiv 0 \pmod{8}$, so $a = (p + 7)/4$ is even — the class of 2 (a residue mod 7) is always present;
- (iii) the factor classes multiply to $a \equiv 4^{-1}p \pmod{7}$.

Enumerating all class multisets with multiplicities capped at 3 subject to (i)–(iii) yields 1536 consistent configurations; for each, one checks whether the target class is attained by a bounded-exponent subproduct. The verification (code: `theory.verify_R7_finite`) confirms: the target is attainable precisely when some factor class is a non-residue. $\square$

Remark 3.1. The hard-class hypotheses are essential: for $p = 701 \equiv 1 \pmod{4}$ (not hard), $a = 177 = 3 \cdot 59$ has the non-residue factor 3, yet $R = 7$ fails. Constraint (ii) — the forced even factor — is exactly what rescues every would-be Type II configuration at hard primes. Empirically (Section 5), among 41,421 sampled failures of $R = 7$ at hard primes, *all* are Type I.

3.4 Proof of Theorem 1.3

By Theorem 1.4 with $R = 11$, $d = 10$: multiplicities cap at 5, and at hard primes $3 \mid a$ (from (H2), $11 \equiv 2 \pmod{3}$), while $2 \nmid a$ (from (H1), $p + 11 \equiv 4 \pmod{8}$) and $7 \nmid a$ (from (H3), $-11 \equiv 3 \pmod{7} \notin \{1, 2, 4\}$). A dynamic-programming enumeration over the finite configuration space (code: `theory.finite_criterion_dp`) compresses it to 25 equivalence states: 16 succeed, 6 fail with all factor classes quadratic residues (case (a)), and exactly 3 fail with a non-residue present (case (b)); the three failing states decode to the three listed patterns. Two structural facts emerge from the enumeration. First, no state fails by a *proper-subgroup* obstruction: the only subgroup of $\mathbb{Z}/10$ containing an odd element properly is $\{0, 5\}$, and the consistency relation forces p 's class to extend it to all of $\mathbb{Z}/10$. Second, the case-(b) failures are pure exponent-budget edge cases: any of $v_3(a) \geq 2$, a factor in class $\{4, 5, 9\}$, or a second non-residue factor restores solvability. The criterion agrees with ground-truth divisor search for 158,759 of 158,759 hard primes tested below 10^9 . $\square$

4 The sieve bound

Proof of Theorem 1.5. Step 1 (exact criteria). Let p be a hard prime and set $n = (p + 3)/4$, so that $(p + 7)/4 = n + 1$: the two shifted forms are consecutive integers. By Theorems 1.1 and 1.2, p is exceptional (both residuals fail) precisely when

$$n \text{ has no prime factor } \equiv 2 \pmod{3} \quad \text{and} \quad n + 1 \text{ has no prime factor } \equiv 3, 5, 6 \pmod{7},$$

with $p = 4n - 3$ prime.

Step 2 (congruence bookkeeping). Fix one of the six hard classes $h \pmod{840}$; then $n \equiv (h + 3)/4 \pmod{210}$ is fixed, and the conditions at the primes 2, 3, 5, 7 are determined by the class (in particular $3 \nmid n$ and $7 \nmid n + 1$ automatically). It suffices to bound each class and sum.

Step 3 (containment in a sifted set). Let $z = x^{1/10}$. After discarding the $O(z)$ primes $p \leq z$, every exceptional p gives $n \leq (x + 3)/4$, $n \equiv n_0 \pmod{210}$, such that for every prime q with $7 < q \leq z$:

- (a) $q \equiv 2 \pmod{3} \implies n \not\equiv 0 \pmod{q}$;
- (b) $q \equiv 3, 5, 6 \pmod{7} \implies n \not\equiv -1 \pmod{q}$;
- (c) $q \nmid 4n - 3$ (since $4n - 3 = p$ is prime, $p > z$).

Conditions (a), (b) only use “no forbidden factor up to z ”, which is implied by the full criteria — an upper bound needs no more.

Step 4 (sieve of dimension 2). This is a standard upper-bound sieve for the three linear forms n , $n + 1$, $4n - 3$, with sifting function

$$\omega(q) = \mathbf{1}_{q \equiv 2(3)} + \mathbf{1}_{q \text{NQR}(7)} + 1 \quad (q > 7),$$

whose forbidden classes $0, -1, 3 \cdot 4^{-1} \pmod{q}$ are pairwise distinct for $q \nmid 21$. By Dirichlet, ω has average $\kappa = \frac{1}{2} + \frac{1}{2} + 1 = 2$ in the Halberstam–Richert sense $\sum_{q \leq w} \omega(q) \log q/q = \kappa \log w + O(1)$. The Fundamental Lemma of sieve theory ([3, Thm. 2.5]; [2, Lem. 6.3]) with $z = x^{1/10}$ gives

$$\#E(x) \ll x \prod_{7 < q \leq z} \left(1 - \frac{\omega(q)}{q}\right) \asymp x (\log z)^{-1} (\log z)^{-1/2} (\log z)^{-1/2} \asymp \frac{x}{(\log x)^2},$$

by Mertens’ theorem and its arithmetic-progression form. Summing over the six classes preserves the bound.

For the second assertion, Theorem 1.3 adds the form $(p + 11)/4 = n + 2$ — *three consecutive integers* — with the case-(a) condition “no prime factor among the five non-residue classes mod 11” (dimension $\frac{1}{2}$); the case-(b) configurations demand all factors of $n + 2$ within 4 of the 10 unit classes, a condition of strictly higher dimension ($\frac{6}{10}$), hence negligible. Total dimension $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + 1 = \frac{5}{2}$. $\square$

Proof of Theorem 1.6 (sketch). By Proposition 2.2, failure at R implies $(p + R)/4$ has no prime factor in at least one explicit class mod R , a sifting condition of dimension $\geq 1/\varphi(R)$ on the form $(p + R)/4$; distinct R give distinct shifts. Since $\sum_{R \equiv 3(4)} 1/\varphi(R)$ diverges, a finite set $S(A)$ achieves total dimension $> A$, and the multidimensional Selberg sieve yields the claim. $\square$

Remark 4.1. Theorem 1.6 is the strongest statement the pure sieve can give: density $O((\log x)^{-A})$ for every A , but not emptiness. The conjecture for hard primes is equivalent to a *finite covering hypothesis*: some finite S admits, for every hard prime, a residual $R \in S$ with (3). All computational evidence below supports $S = \{3, 7, 11, \dots, 107\}$.

5 Computations to 10^{10}

All code and data are in the accompanying repository. Certificates are found by direct integer factorization of a (trial division; the factorization of $m^2 = p^2 a^2$ follows) and scanning divisors for the class condition (3); every certificate is re-verified in exact integer arithmetic, $4abc = n(bc + ac + ab)$.

5.1 Data

Every one of the 14,215,707 hard primes $p < 10^{10}$ possesses a residual certificate with $R \leq 107$. The minimal residuals distribute as follows (10^{10} data):

R	count	R	count	R	count
3	7 386 405	31	30 222	63	57
7	3 472 188	35	4 775	67	18
11	2 325 588	39	4 307	71	46
15	487 889	43	1 131	75	3
19	322 871	47	1 661	79	11
23	150 828	51	240	83	5
27	27 000	55	263	107	1
		59	198		

Salient features:

- $R_{\max} = 107$, attained by the single prime $8,803,369 < 10^7$, *unchanged from 5×10^7 to 10^{10}* — three orders of magnitude.
- No prime has minimal $R \in \{87, 91, 95, 99, 103\}$. This gap requires no modular explanation: under the calibrated independence model the expected counts there are 0.12, 0.03, 0.05, 0.01, 0.01. Conditioned on failing every $R \leq 83$, a prime lands at 87 with probability 0.53 and at 107 with probability 0.017; the record prime is the anomaly, not the gap.
- Among all 1,587,581 hard primes below 10^9 , exactly *four* admit a unique working residual ≤ 107 (8,803,369 with 107; 142,361,209 with 59; 287,567,281 with 83; 794,037,841 with 63) — the entire obstruction to a shorter covering list. A greedy cover of the full solvability table uses 18 residuals; a packing argument gives a lower bound of 12.

5.2 Calibration of Theorem 1.5

The proportion of hard primes failing both $R = 3$ and $R = 7$, multiplied by $\log x$, is constant to within 2% across three decades of the 10^9 data:

bin ($p \sim$)	10^6	10^7	6×10^7	2.5×10^8	10^9
density $\times \log x$	5.14	5.19	5.18	5.17	5.16

That is, the joint exceptional density is $\approx 5.17/\log x$: the sieve exponent of Theorem 1.5 is exactly right, and 5.17 estimates the (conjectural) Selberg–Delange constant. The exceptional set is density zero but decays slowly: 26.5% of hard primes below 10^9 fail both residuals; all are settled by deeper residuals of the list. The fitted per-residual decay exponents likewise match the sieve prediction $\kappa = \frac{1}{2}$ where exact criteria exist (e.g. $\hat{\kappa} = 0.539$ for $R = 7$).

5.3 Failure taxonomy

Sampling the complete solvability table at 10^9 : failures at $R = 7$ are 100% Type I (as Theorem 1.2 requires); at $R = 11$, 89.7% Type I and 10.3% the budget pattern of Theorem 1.3(b); the Type I share decreases with $\varphi(R)$ (e.g. 51% at $R = 103$), as hitting one specific class among $\varphi(R)$ grows harder while the character dichotomy stays binary.

6 Open problems

1. **Finite covering.** Prove that some finite admissible set S covers all hard primes (Theorem 1.6 gives density $(\log x)^{-A}$ for every A ; emptiness is beyond the sieve).
2. **More exact criteria.** The meta-theorem makes every fixed R decidable; $R = 15$ (first composite, coupled conditions mod 3 and 5) and $R = 19, 23$ are natural next cases, each raising the proven exponent in the chain of Theorem 1.5.
3. **The record prime.** 8,803,369 fails every residual below 107 even allowing $R \leq 400$; its shifted values are anomalously smooth (at $R = 107$, $a = 3^2 \cdot 11^2 \cdot 43 \cdot 47$). A quantitative theory of joint failure for smooth-shift primes would explain why the record is static and bound its future growth.

References

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